

Part 0 — Minimal background before EpiProfile (absolute beginner mode)

This section is intentionally basic. The goal is that any reader (even without prior LC-MS experience) understands what RT, m/z , “theoretical mass”, and “ppm tolerance” mean, and why a tool like EpiProfile may output zeros when one of these assumptions breaks. If you are comfortable with LC-MS, you can jump directly to Part 1.

0.1 What LC-MS measures (one sentence)

In bottom-up proteomics, proteins are digested into peptides. Liquid chromatography (LC) separates peptides over time (each ion elutes at a retention time, RT). Mass spectrometry (MS) measures their mass-to-charge ratio (m/z) and, when fragmented (MS2), provides evidence to confirm identity and modifications. For XIC-based quantification, the core question is: “how much signal is present near this m/z within this RT window”.

0.2 Retention time (RT): definition, drift, and search windows

RT is the time coordinate where an ion (a peptide at a given charge state and modification state) appears as a chromatographic peak. RT is driven primarily by the LC setup (column chemistry, gradient, flow rate, temperature, dead volumes). Therefore, across datasets, instruments, or LC methods, the same peptide can shift by minutes. Many pipelines assume shifts are small and use reference RTs to “guide” the search; if the reference comes from a different chromatography, it can be worse than having no reference at all.

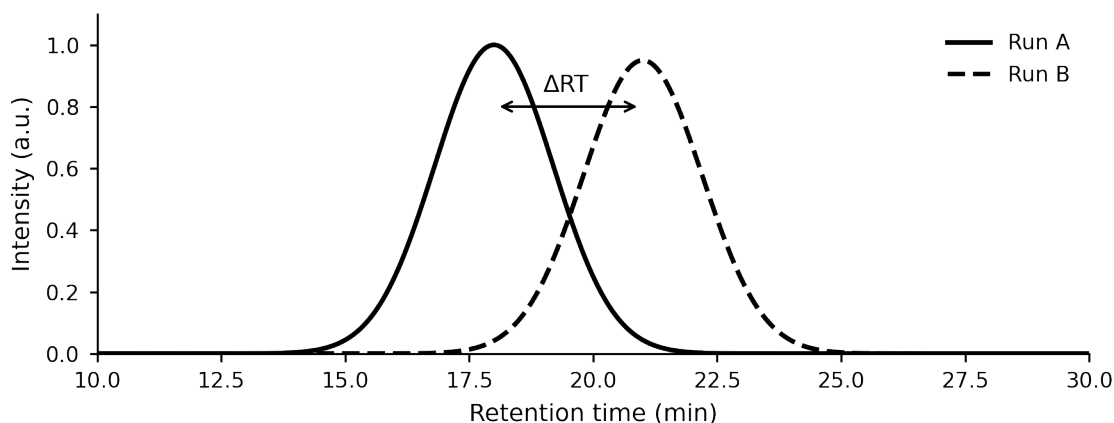


Figure 0.1. Conceptual RT drift: the same peptide can elute minutes later in another run/dataset.

Rule of thumb: RT reference is only trustworthy when RAW files share the same LC/method and are temporally close. If you mix platforms (e.g., ZenoTOF vs Orbitrap) or different LC methods, split runs into homogeneous groups or run without global RT reference.

Key idea: retention time (RT) is not an intrinsic property of a peptide; it is a property of the LC system for that specific run. RT is driven primarily by the column (chemistry/length/ID), the gradient, flow, temperature, dwell volume, and column aging. Therefore, across days/weeks—or across platforms (e.g., an Evosep/ZenoTOF setup versus an Orbitrap with a different LC)—the

same peptide can shift by multiple minutes, and the shift is not always a rigid uniform offset for all peptides.

In EpiProfile, RT_ref is used as a hypothesis to guide detection: it is translated into RT search windows around an expected elution time. When RT_ref is portable (same LC method, runs close in time), a narrow window increases sensitivity and reduces false positives. When RT_ref comes from a different chromatography, the guided window may miss the true peak and the typical output becomes 'not detected' → intensity=0 and RT=0, even though the peptide is present. A sudden explosion of zeros (especially in unmodified anchors) is the single most diagnostic symptom of an inappropriate RT reference.

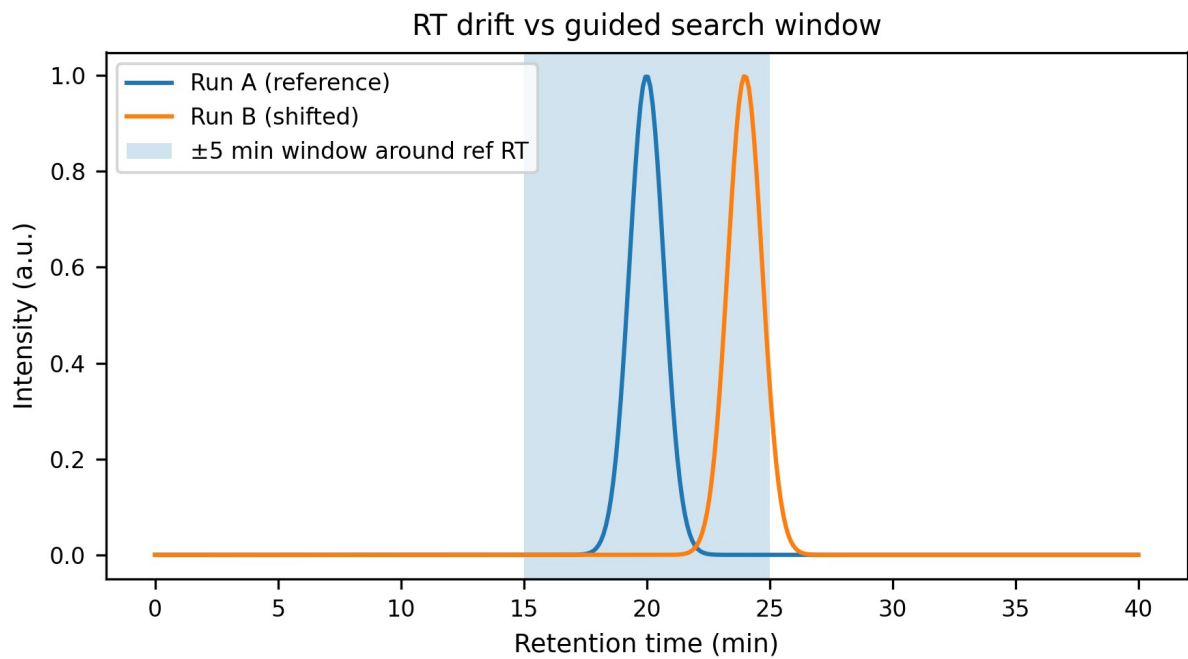


Figure 0.2b. Schematic example: RT drift between runs and a ± 5 min guided window around RT_ref.

0.3 Mass, charge, and m/z: how MS1 represents a peptide

MS does not measure neutral mass directly; it measures m/z. A given peptide can be observed as multiple ions ($z=1,2,3,\dots$) depending on how many charges (protons) it carries. The relationship is:

$$m/z = (M + z \cdot H) / z$$

where:

M = monoisotopic peptide mass (Da)

z = charge state

H = proton mass ≈ 1.007276 Da

[1]

Each ion also shows an isotopic envelope: peaks separated by $\sim 1/z$ in m/z. This spacing helps infer charge and distinguish true signal from noise.

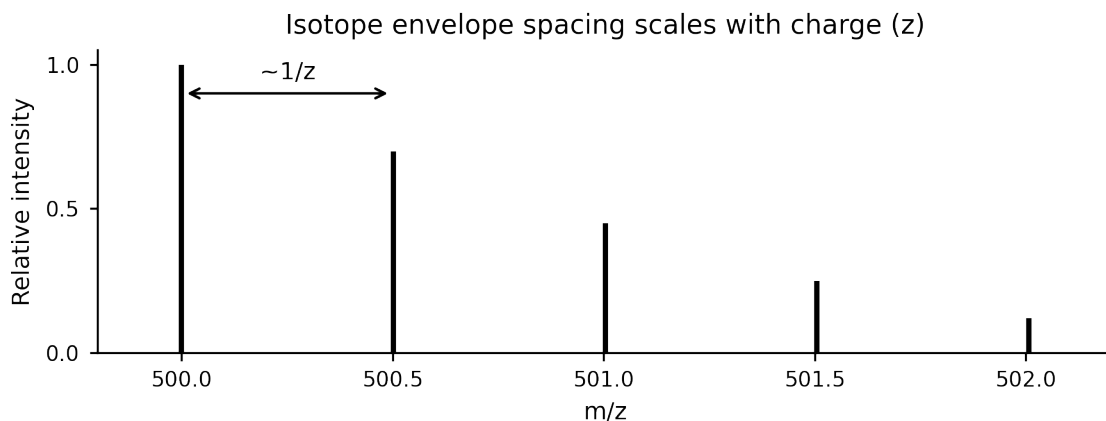


Figure 0.2. Isotopic envelope: peak spacing is approximately $1/z$ (here, $z=2$).

MS does not directly weigh neutral molecules; it measures m/z of ions. The same peptide can be observed as $1+$, $2+$, $3+$, etc., and each charge state yields a different m/z . EpiProfile encodes this explicitly via `pep_ch` and `pep_mz` (one row per peptideform, one column per charge). Practically, a single chemical species often produces multiple candidate XIC traces, and assignments are more robust when charge states cohere (same RT, consistent isotopic envelope).

Useful intuition: the spacing between adjacent isotopes in MS1 is $\sim 1/z$ in m/z units. At higher charge, the isotopic envelope becomes compressed along the m/z axis. This helps distinguish real signals from interferences, but also increases the demands on mass accuracy when many near-isobaric forms are present.

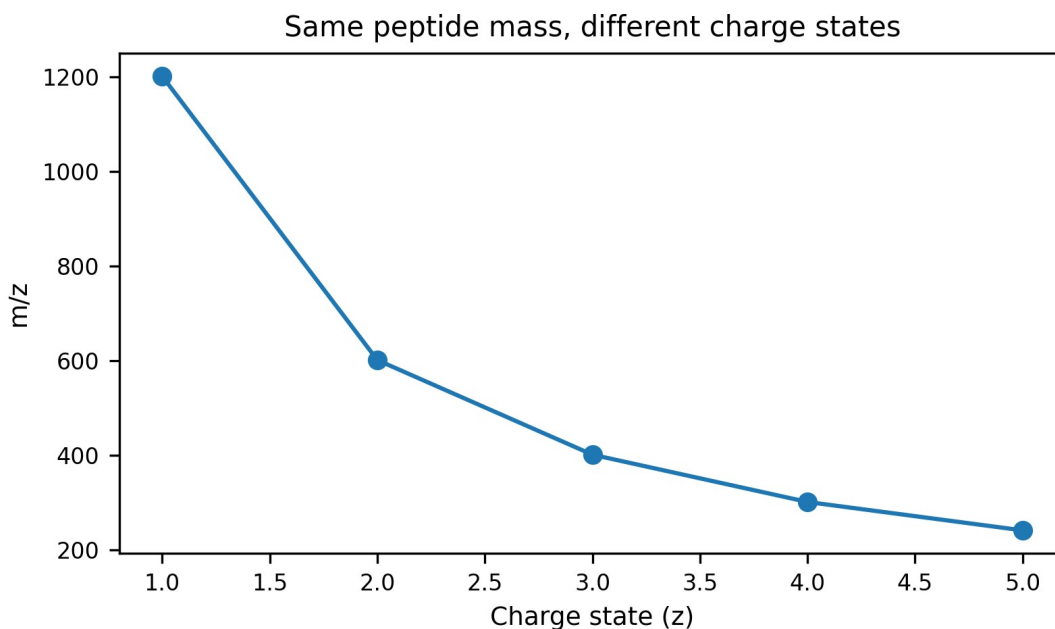


Figure 0.3b. Same monoisotopic mass, different charge states $\rightarrow$ different m/z values.

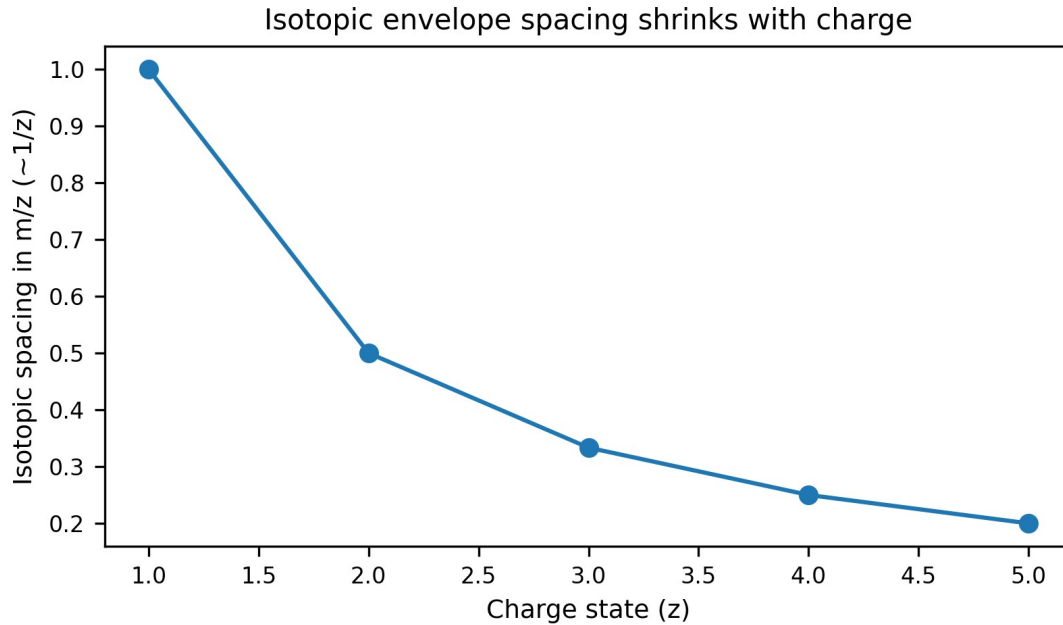


Figure 0.3c. Approximate isotopic spacing ($\sim 1/z$): decreases with higher charge.

0.4 Theoretical mass, modifications, and ppm tolerance

Theoretical mass is computed from the peptide sequence (sum monoisotopic amino-acid masses + water) and then adjusted by modification mass deltas. Examples (approx.): me1 +14.0157, me2 +28.0313, me3 +42.0470, ac +42.0106. In histone workflows with propionylation, propionyl adds $\sim +56.0262$ to the peptide N-terminus and to unmodified lysines (functional consequence: trypsin behaves Arg-C-like).

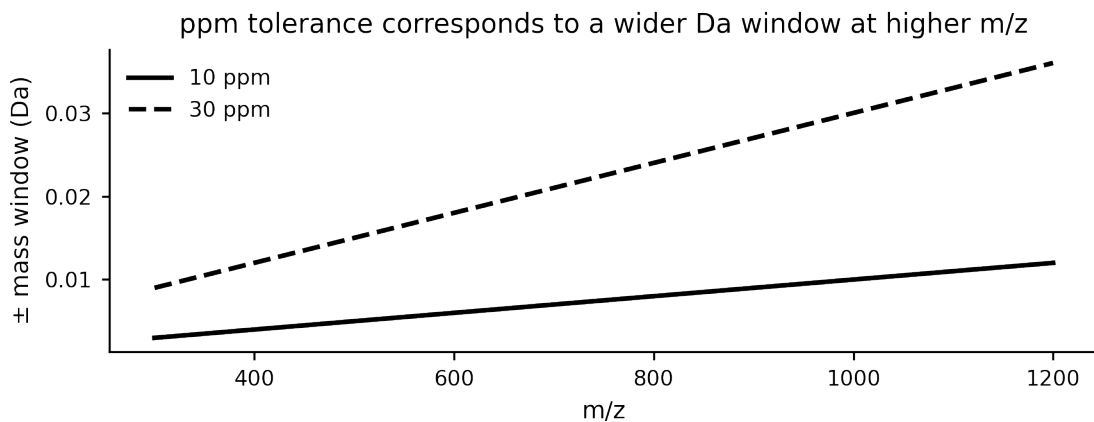


Figure 0.3. A ppm tolerance corresponds to different Da windows depending on m/z (so 10 ppm is not a fixed Da).

Observed vs theoretical matching is decided using a tolerance, commonly expressed in ppm:

$$\text{ppm_error} = 1e6 \cdot (m/z_{\text{observed}} - m/z_{\text{theoretical}}) / (m/z_{\text{theoretical}})$$

Quick intuition:

at $m/z = 500$, 10 ppm $\approx \pm 0.005$ Da
at $m/z = 1000$, 10 ppm $\approx \pm 0.010$ Da

Increasing tolerance (e.g., 20–30 ppm) increases sensitivity when there is systematic mass error, but also increases the risk of incorrect matches—especially in histones, where many peptide families are highly similar. Start conservative (10 ppm) and only relax with evidence.

‘ppm’ is a relative error. In histone analyses (many near-isobaric peptide forms), p_{tol} sets a trade-off: too strict and you lose sensitivity under calibration drift; too loose and you increase the risk of matching the wrong peak (especially in regions with multiple modified forms and partial co-elution).

Quick mental conversion: $\Delta Da \approx (m/z) \times (\text{ppm}) \times 10^{-6}$. For example, at $m/z=600$, 10 ppm $\approx \pm 0.006$ Da. This linear scaling explains why the absolute Da window increases with m/z even when the ppm setting is constant.

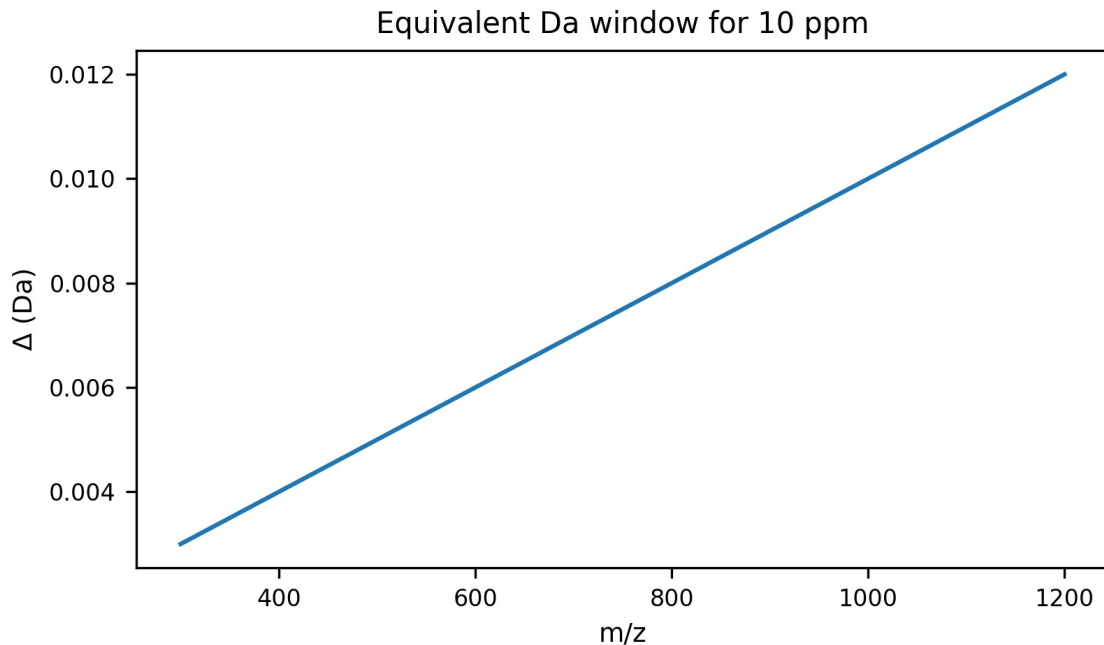


Figure 0.4b. Absolute Da window equivalent to 10 ppm across a range of m/z values.

$$\text{ppm_error} = (\text{mz_observed} - \text{mz_theoretical}) / \text{mz_theoretical} * 1\text{e}6$$

0.5 XIC/EIC and area-based quantification (AUC)

In LC–MS proteomics, the most direct quantitative observable is intensity integrated over time. This is why most bottom-up quantification workflows converge on extracted chromatograms: XIC/EIC (extracted ion chromatogram / extracted ion current). Conceptually, an XIC is simply the MS1 intensity for a target m/z tracked along the LC retention time axis. [1,2]

Operationally, an XIC is built by summing intensities within a mass window (target $m/z \pm \text{ppm}$ tolerance), often aggregating multiple isotopes and, when relevant, multiple charge states. The area under the chromatographic peak (AUC) provides a robust proxy for relative abundance, provided that the peak is correctly bounded (RT window) and the target ion is well-defined (theoretical mass, charge, isotope set, and potential interferences). [1,2]

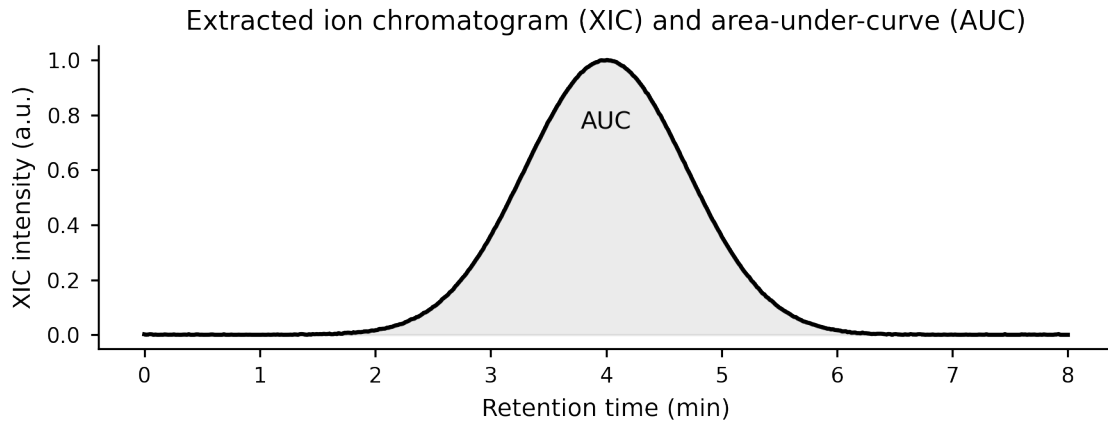


Figure 0.4. Minimal schematic of an XIC and its area-under-curve (AUC) quantification.

For histones, this point is amplified because many peptideforms are isobaric or near-isobaric (PTM/derivatization combinations with very similar masses). EpiProfile relies on layouts and RT-guided windows specifically to reduce the chance of integrating the “wrong peak” when several nearby signals coexist in the chromatogram. [3,4]

```
# Conceptual AUC (not an implementation)
#  $XIC(t) = \sum_{m/z \text{ in window}} \text{intensity}(m/z, t)$ 
#  $AUC = \int_{t_1}^{t_2} XIC(t) dt$ 
```

In MS1-based label-free quantification, the practical signal is the area under the extracted ion chromatogram peak (XIC/AUC). An XIC is built by filtering MS1 within a narrow m/z band (ptol) and tracking intensity over RT. Integrating the peak (AUC) is typically more stable than using a single apex intensity because it averages over noise.

The common failure point is not AUC arithmetic but peak definition: baseline, co-elution, and interference. This is why EpiProfile combines RT constraints (windows), isotopic logic, and—when available—MS2 evidence (special.nDAmode) to reduce mis-assignments.

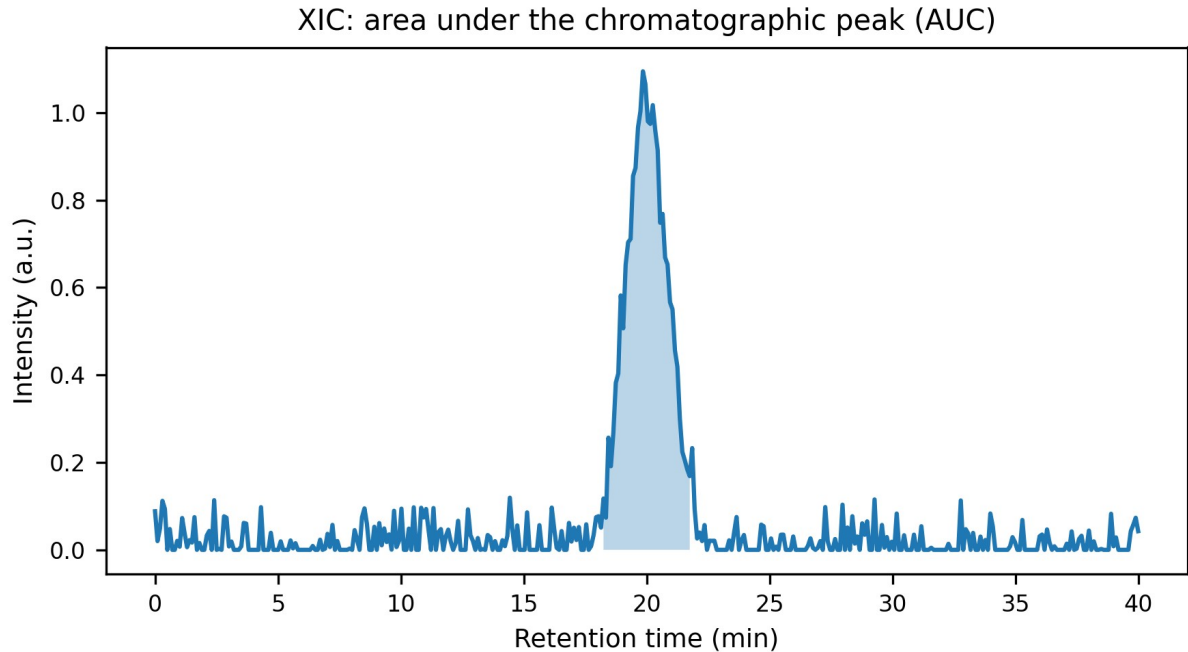


Figure 0.5b. Schematic XIC with the AUC (shaded) used for quantification.

0.6 Minimal EpiProfile nomenclature (hDP, hPF, PTMs, and mod_type)

EpiProfile operates at three layers: (i) histone-derived peptides (hDPs) defined by sequence/region (e.g., H3_3_8), (ii) peptideforms (hPFs), which are concrete PTM combinations on that peptide (e.g., K4me3, K4ac), and (iii) PTM summaries (hPTMs) that aggregate peptideforms sharing a mark. In MATLAB, each peptideform is defined by two labels:

```
His.mod_short = {'unmod'; 'K4me1'; 'K4me2'; 'K4me3'; 'K4ac'}
His.mod_type = {'0,pr;2,pr'; '0,pr;2,me1'; ... }
```

Practical interpretation:

- '0,pr' : modification at peptide N-terminus (position 0)
- '2,me2' : modification at peptide position 2 (1-based indexing), here the lysine in TKQTAR
- suffix (pr, me1/2/3, ac, ...) encodes the modification type

This is useful because many modules treat the unmodified form as an anchor: once unmod RT is detected, expected RTs for other peptideforms are shifted by a simple delta within that run.

0.7 Minimal checklist before running (and before going to R)

- 1) Decide the RT-reference scope: per dataset/homogeneous group (recommended) or no global reference.
- 2) Isolate outputs: one output per RAW (or at least per dataset); never reuse outputs across instruments.
- 3) Change ptol (ppm) only with evidence of mass error.

4) If many zeros appear: first rule out reference contamination (0_ref_info.mat / shared outputs) before interpreting biology.

0.8 From theoretical mass to search windows: the bridge to EpiProfile

Before discussing `rt_ref` or layouts, it helps to lock in the minimal mental pipeline: a peptide sequence (plus PTMs/derivatization) defines a theoretical monoisotopic mass; that mass, together with a charge state z , defines a theoretical m/z ; finally, real signals are sought near that m/z within a tolerance (ppm) and within a plausible RT window. [1,2]

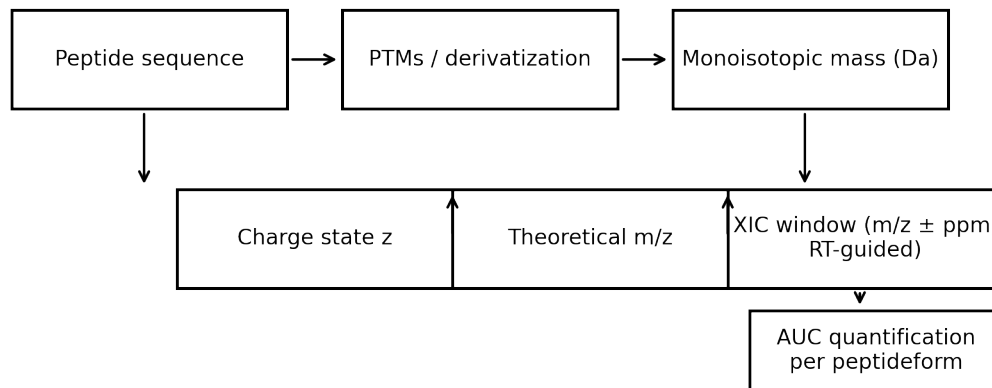


Figure 0.5. From peptide theory to AUC quantification through m/z and RT windows.

Operationally, EpiProfile does not attempt a de novo peak search across the full chromatogram. Instead, it starts from a closed set of targets (peptides/peptideforms) with theoretical masses and approximate RTs, and applies an RT-guided extraction to build XICs and integrate areas. This design improves reproducibility at cohort scale, but it also explains sensitivity to incompatible RT references: if `rt_ref` is contaminated or mismatched to the current LC-MS setup, the guided window can miss the true peak and produce $AUC \approx 0$. [2,3]

```
# Mass → m/z (monoisotopic; approximation)
M = M_peptide + ΣΔM_mods + M_H2O
m/z = (M + z·mH) / z    # mH≈1.007276 Da
Δm (Da) = (ppm/1e6) · (m/z)
```

EpiProfile — Retention-Time (RT) References: Conservative Operation & Failure-Mode Guide

A black-on-white primer + troubleshooting playbook for non-experts (safe across Orbitrap / ZenoTOF / mixed LC setups)

Version: v1.10-en (whitepaper, Nature-style figures; metro maps only for conceptual workflows)

Table of Contents (manual)

This document is structured with numbered headings; in Word you can insert an automatic TOC via References → Table of Contents.

Primer (for bioinformatics / wet-lab readers)

This page intentionally starts from first principles. It is meant for people who can run scripts but do not routinely debug LC-MS quantification.

EpiProfile is a targeted, layout-driven quantification pipeline for histone-derived peptides (propionylation + trypsin/Arg-C-like digestion) where many peptideforms are close in mass and can co-elute. It does not try to solve full de novo identification; it tries to measure a curated panel robustly.

Most EpiProfile failures that end as “zeros” are not numerical bugs. They are upstream failures to locate a peak (wrong m/z window, wrong RT window, missing scans, or wrong reference being imposed).

0.9 Conservative 5-minute operational checklist (before running)

This section condenses the minimum operational checks that prevent the most expensive failure modes: (i) applying RT references learned on one instrument/LC method to a different one, (ii) forcing a narrow guided window that contains no scans/peaks, and (iii) leaking intermediate outputs (e.g., *.xls) across RAWs when output folders are shared.

Decision	Conservative choice	When NOT to use	No-code mitigation
Reference RT (ndebug)	ndebug=0 + homogeneous folders	Mixed instruments/LC/methods or weeks/months apart	Split raw_path by dataset/homogeneous group; delete histone_layouts/0_ref_info.mat when reusing a folder.
No reference	ndebug=2	When you need maximum sensitivity for rare PTMs and the dataset is truly homogeneous	Avoids cross-dataset reference; compensate with RT/zero-rate QC.
Mass tolerance (ptol)	ptol=10 ppm	Systematic mass shift or poor calibration	Increase to 20–30 ppm only if a consistent shift is observed; document the rationale.
Intermediate outputs	unique outpath per RAW (or per dataset)	Multiple RAWs writing to the same output folder	Use per-RAW subfolders; avoid H3_04v3_*.xls from another run.

raw_path/ (one folder)

histone_layouts/0_ref_info.mat → RTs learned from a subset of RAWs

If you mix instruments/methods here:

- the same RT template is applied across all RAWs
- 'guided' windows may miss peaks → many zeros

Conservative layout:

dataset_A/raw_path_A (own 0_ref_info.mat)

dataset_B/raw_path_B (own 0_ref_info.mat)

Figure 0.9a | Scope of RT references: if histone_layouts/0_ref_info.mat exists, RT reference is folder-global (raw_path). Conservative design: one folder per dataset/homogeneous group.

Minimal folder template (conservative recommendation):

```
PXD014739_orbitrap/  
raw/           # *.raw (homogeneous)  
histone_layouts/ # created during the run  
output/        # optional: per-RAW subfolders  
PXD046034_orbitrap/  
raw/  
histone_layouts/  
output/
```

Key vocabulary (minimal but precise)

- RAW: vendor file containing MS1 and (optionally) MS2 scans (e.g., Orbitrap *.RAW).
- MS1: full-scan spectra used to build XIC/EIC traces; EpiProfile integrates an area-under-curve (AUC).
- MS2: fragmentation used to validate and disambiguate near-isobaric peptideforms when enabled (special.nDAmode).
- RT (retention time): minute at which a peptide elutes; controlled mostly by LC conditions (column, gradient, flow, temp), not by the mass analyzer itself.
- ppm (mass tolerance): m/z acceptance window around the theoretical value; too tight → false negatives, too loose → false positives.
- unmod: the "unmodified" peptideform used as an intra-run anchor; if unmod is missing, the whole RT logic becomes fragile.
- raw_path: the input folder; some references are global per raw_path (critical).

- `outpath / cur_outpath`: output folder; some modules read other modules' *.xls outputs from `outpath` (also critical).
- `ndebug`: mode flag: 0=normal+reference RT, 1=debug, 2=normal but no reference RT (per-file processing without `0_ref_info`).

NOTE: Why “Orbitrap vs ZenoTOF” is mainly an LC question

Retention time is dominated by LC (column+gradient). Switching platforms often implies different LC methods. Therefore, reusing RT references across datasets is high-risk unless the LC setup is demonstrably comparable.

- In non-model species, identification relies even more on MS2 quality and database completeness; targets must be adapted.
- Global PTM landscapes require controlling digestion/derivatization biases and careful normalization.
- Operational takeaway for EpiProfile: the further you are from the method it was optimized for, the more conservative you should be with RT references and mass tolerances.

D) Non-model tissues strategy: robustness when the catalogue is incomplete

- Pre-digestion propionylation blocks lysines and enforces Arg-C-like digestion.
- O-propionylation (Ser/Thr/Tyr) must be managed because it disperses signal; some protocols include a high-pH “reversal” step.
- Isobaric ambiguities exist: without MS2 and strict rules, peptideforms can be misassigned.
- Histone variants may differ by a single amino acid: mass and RT change; curated variant lists (as in EpiProfile_PLANTS) become essential.

C) Arabidopsis histone MS/MS chapter: why propionylate and how it breaks

- Every stage can introduce bias: extraction → reduction/alkylation → digestion → desalting → LC → acquisition → database search.
- DDA has sampling bias (Top-N) and missingness; DIA reduces missingness but changes assumptions.
- Identification uses FDR control; MS1 quantification is sensitive to co-elution and co-isolation.
- Parameters like mass tolerance propagate into both ID and quantification quality.

B) Bottom-up overview (where artefacts originate)

- Instrument = ionization (ESI) + analyzer (e.g., Orbitrap/TOF) + detector; the analyzer largely determines resolution and accuracy.
- Observed m/z depends on charge state (z): the same species can appear at multiple z .
- Isotope envelope: EpiProfile integrates mono-isotopic and/or summed isotopes; poor centroiding/export can reduce sensitivity.
- RT is a physical LC coordinate, not “software”: it changes with column/gradient/temperature.

A) General MS-proteomics intro (concepts that later reappear as errors)

These notes synthesize four reference documents (general MS-proteomics introduction, a bottom-up overview, and two histone-focused protocols/strategies). They are not literal summaries; they are the points that directly affect how you configure and debug EpiProfile.

0.5.8 Appendix: distilled “gold” from four readings (for the white paper)

RAW level:

- Total gradient length (30/60/90+ min) and MS1 scan density.
- Stable base signal (spray) and no extreme saturation.
- If rare PTMs are expected, ensure DDA settings are not too aggressive (dynamic exclusion, Top-N) for low-abundance forms.

Dataset level (PXD / instrument / LC method):

- Ensure homogeneity: same instrument + same column + comparable gradient.
- Decide RT strategy: reference per homogeneous group (recommended) or no reference.
- Validate mzML conversion: MS1 and MS2 present, RT units consistent, centroiding correct.
- Confirm histone signal exists (at least a few abundant unmodified landmarks).

0.5.7 Conservative “before you press run” checklist

Prep failures that manifest as mass/RT problems: incomplete propionylation (extra species), over-propionylation on Ser/Thr/Tyr (signal dispersion), losses during cleanup, and isobaric ambiguities (e.g., modifications or combinations with identical nominal mass).

For histones, propionylation blocks free lysines to (i) reduce charge heterogeneity and (ii) make trypsin behave Arg-C-like (cleaving after Arg). This stabilizes N-terminal peptides and improves quantification. Commonly there are two rounds: propionylation before digestion (lysines) and after digestion (N-termini).

0.5.6 Histones: why sample prep changes the rules

In DDA, MS2 is only triggered for selected precursors; in complex mixtures or low abundance targets, a precursor may not be sampled. That is why EpiProfile quantifies from MS1—but that makes accurate RT and m/z control essential.

DDA and missingness: missing MS2 does not mean missing peptide

EpiProfile uses RT references as a shortcut to accelerate and disambiguate. When the reference is compatible, it helps. When it is not, it can fail silently (zeros) even if the peptide exists.

RT is determined by LC (column chemistry, gradient, temperature, dwell volume, flow). Switching from Orbitrap to TOF often implies LC/method changes, and the “2–3 min shift” assumption may fail. Therefore, an RT reference learned on another column/method can be actively harmful: it guides the search to empty windows and yields zeros.

0.5.5 Retention time (RT): mostly chromatography, not the analyzer

Idea (pseudo): if you already have an ID table with mz_theoretical and mz_observed
$$\text{ppm_error} = 1e6 * (\text{mz_observed} - \text{mz_theoretical}) / \text{mz_theoretical}$$

```
summary(ppm_error); hist(ppm_error)
```

Red flag: mean $\neq$ 0 and heavy tails → reconsider def_ptol or mzML conversion choices

If you have an external identification table (Mascot/PD/MaxQuant/Skyline), inspect the ppm error distribution. If not, use a couple of abundant landmarks (including an unmodified histone peptide) and verify whether peaks are consistently shifted.

Quick check (without touching EpiProfile): estimate mass error in an mzML

Practical rule: increase def_ptol only with objective evidence of systematic mass error (consistent shift across several targets or system suitability controls). Raising tolerance “just in case” often converts a sensitivity issue into a specificity issue.

The ppm tolerance defines how far from the theoretical m/z a peak can be and still be considered a candidate. If the instrument is well calibrated and mzML export is correct, 10 ppm is typically conservative. If there is systematic mass error (calibration, centroiding artifacts, odd profiles), you may need 20–30 ppm to recover signal—but the cost is more false positives, especially for histones (dense, near-isobaric signal).

0.5.4 Mass tolerance (ppm): why “10 ppm” is not magic

MS2 (fragmentation) is used to confirm identity: you isolate a precursor (m/z, z) and fragment it. In DDA this happens for a subset (Top-N), so MS2 can be missing even with strong MS1. In histones, many combinations are isobaric; MS2 (and strict rule-based logic) often determines correct assignment.

MS1 (survey) is a time series of m/z peaks: each peptide appears as an isotope envelope at one or more charge states (z). EpiProfile integrates an extracted ion chromatogram (XIC) area in a defined RT and m/z window.

0.5.3 MS1 vs MS2: what you are actually measuring

Two concepts dominate practical failure modes: (i) mass accuracy and (ii) chromatographic stability (RT). If either breaks, you can obtain many zeros even when the peptide is present.

Bottom-up means: protein → (enzyme) → peptides → LC separates in time → MS measures m/z in MS1 → MS fragments some precursors in MS2 → software assigns sequences/PTMs.

EpiProfile sits at the end: it takes a target list and quantifies MS1 signal (and uses MS2 only as support in ambiguous cases / depending on mode).

0.5.2 The bottom-up proteomics map in 90 seconds

EpiProfile is NOT a general discovery/identification engine: it will not discover peptides de novo, it will not rescue a poorly calibrated LC-MS system, it will not fix an incomplete mzML, and it cannot “guess” RT when the chromatogram is incompatible. Its strength comes from constraining the problem: few targets, explicit modification rules, and reproducible quantification.

EpiProfile quantifies (mostly from MS1) a curated set of histone peptides and their peptideforms (PTM combinations) using XIC/AUC guided by theoretical m/z and expected RT. It assumes you already have valid MS1/MS2 spectra (properly converted) and that the experiment follows a bottom-up workflow (proteins digested into peptides).

0.5.1 What EpiProfile does—and what it does NOT do

This page is intentionally “dummy-bioinfo”: if you have never done LC-MS/MS proteomics, it tells you what the files actually contain and why EpiProfile depends so heavily on mass tolerances and retention times (RT). It is not a full course; it is the minimum context needed to avoid treating zeros (or missing IDs) as biology.

Page 0.5 — Before EpiProfile: the absolute minimum you must understand (bottom-up proteomics 101)

If you retain only one idea from this preface: RT reference is an experimental assumption, not a software property. When the assumption holds, it accelerates and stabilizes; when it does not, it produces zeros. The next sections formalize how to detect this and how to operate conservatively (homogeneous folders, deleting references, or `ndebug=2`).

0.5.12 Direct bridge to the RT section of this document

Symptoms: broad/double peaks; the same target shows two RT modes; reduced reproducibility. Causes: LC issues (gradient, aging column, valves), carryover, or mixing methods in the same folder.

What to check: TIC/BPC, gradient stability, and whether samples truly share the same LC method.

Scenario 4 — RT drift within the same dataset (broad tails, double peaks)

Symptoms: extremely large intensities but inconsistent ratios; flipping between isobaric peptideforms; mutually exclusive marks appear together.

Causes: tolerance too wide in dense regions; weak include/exclude rules; insufficient MS2 support for ambiguities; unexpected chemistry (partial propionylation, O-propionylation).

What to check: (i) review overlapping XICs, (ii) see whether MS2 supports the call when present, (iii) lower ambition—quantify only robust marks first.

Scenario 3 — Signal exists but assignment is wrong (weird/unstable ratios)

Symptoms: some regions quantify well while others are zero; often overlaps regions with relocate or learned RTs.

Causes: reference contamination (folder/outpath reuse), region-specific RT drift, or co-elution/isobaricity pushing the correct peak outside the window.

What to check: in the per-region detail output, whether the search window is reasonable and whether candidate peaks exist nearby.

Scenario 2 — Selective zeros (only some regions fail, e.g., H3_27_40)

Symptoms: ratio matrices are near-zero everywhere; very few identifications; empty XIC plots.
Typical causes: incompatible RT reference guiding empty windows; broken mzML conversion (wrong RT units; missing MS1); def_ptol too strict under real mass error; or simply no histones (failed extraction).

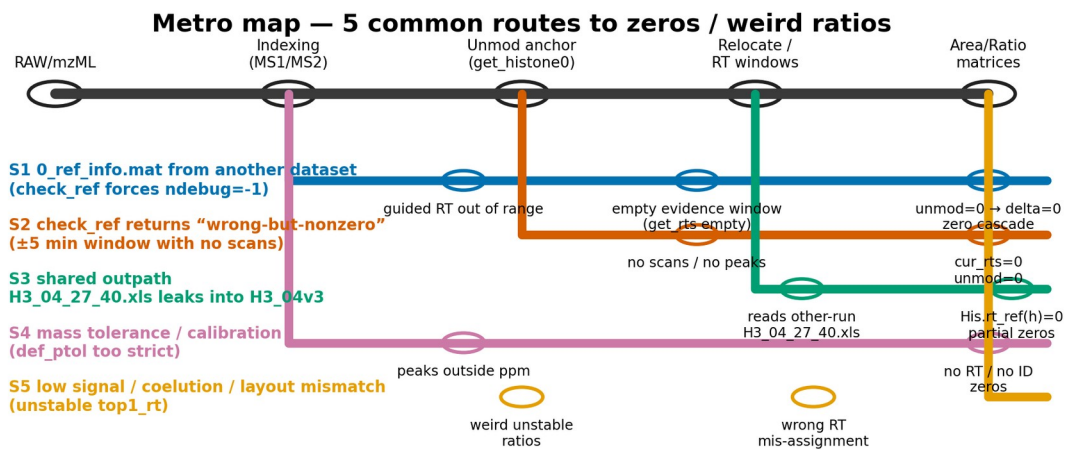
What to check: (i) is a 0_ref_info.mat being reapplied? (ii) total RT range in the mzML, (iii) number of MS1 scans, (iv) a couple of XICs for abundant unmodified histone peptides.

Scenario 1 — Massive global zeros (almost everything 0, including unmodified landmarks)

The goal is not to memorize causes, but to recognize patterns. Below are common symptoms and the minimal verification you can do without touching code.

0.5.11 Catalogue of “pre-EpiProfile” failure modes and how they look in outputs

The following metro-map style diagrams are intentionally used only here: they provide a compact conceptual overview of the most common failure routes and the corresponding mitigations. They are not meant to encode exact software control flow; they are a troubleshooting mental model.



Fast hint: full-zero rows are typically S1-S4; weird ratios often S5 (or S3).

Figure. Conceptual metro-map of common failure routes in EpiProfile RT-guided quantification.

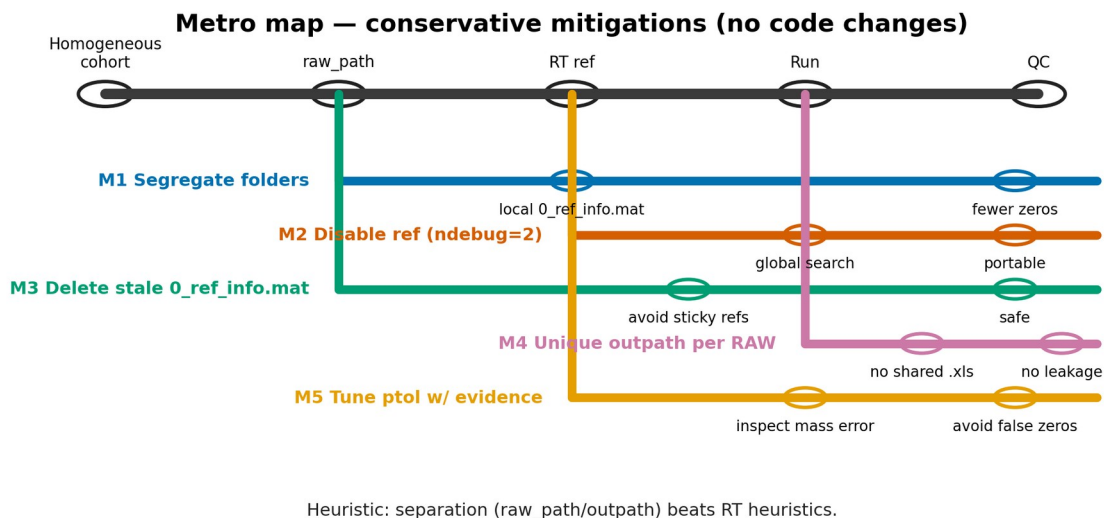


Figure. Conceptual metro-map of practical mitigations corresponding to common failure routes.

Generic example (adapt to your instrument and installation)

```
msconvert input.RAW \
--mzML \
--64 \
--filter "peakPicking true 1-" \
--filter "zeroSamples removeExtra" \
--outfile input.mzML
```

Red flags:

- # - mzML missing MS2 (or MS2 missing precursor m/z/charge)
- # - total RT far from the real LC method
- # - “blocky” peaks / broken isotope envelopes after centroiding

If you use ProteoWizard/msconvert (a common choice), prefer a conservative export: include MS1+MS2, use consistent centroiding, and avoid aggressive filters that change the time grid.

EpiProfile depends on the mzML faithfully representing (i) m/z and (ii) RT. Two avoidable error sources:

- 1) exporting RT in unexpected units or with odd offsets;
- 2) dropping critical information (e.g., MS1 only, or MS2 without precursor metadata) due to conversion options.

0.5.10 RAW → mzML conversion: “minor” details that break RT/mass

MS1 / survey: full scan of m/z (precursors). MS2 / fragmentation: fragment spectrum for an isolated precursor.

Precursor: ion selected for fragmentation. Fragment ions: products of fragmentation.

Centroid vs profile: centroid = peak list; profile = continuous signal. Bad centroiding can distort isotope envelopes.

Resolution (resolving power): ability to separate nearby peaks. Mass accuracy: systematic/random m/z error.

Isolation window: m/z width entering the fragmenter; wide windows increase co-isolation.
Co-isolation / co-elution: multiple peptides enter together → mixed MS2 and ambiguous MS1.
XIC: extracted ion chromatogram (intensity vs RT). AUC: area under the curve (quantification).
PSM: peptide-spectrum match. FDR: false discovery rate (false positives control).
Dynamic exclusion: DDA rule to avoid repeatedly fragmenting the same precursor.
Isobaric: same (nominal) mass for different species; common in histone PTMs.

0.5.9 Operational glossary (the jargon that shows up in logs and discussions)

Mental model: the three RT layers in EpiProfile

Understanding these three layers explains almost every case of unexpected zeros:

Layer 0 — hardcoded RT templates inside each peptide module

Each module defines a His struct with theoretical m/z (per charge) and a hardcoded His.rt_ref vector for all peptideforms. Example: H3_01_3_8 (TKQTAR) defines 5 entries (unmod, K4me1/2/3, K4ac) with fixed RTs.

These templates are not universal physics. They are empirical, tuned to a specific LC context used during development.

Layer 1 — per-folder global reference (0_ref_info.mat) loaded by check_ref()

When ndebug is 0 (default) or -1, check_ref() looks for raw_path/histone_layouts/0_ref_info.mat. If found, it can overwrite His.rt_ref(1) (unmod RT) with a value learned from a specific set of RAWs stored in that MAT.

Crucial: the reference is global per raw_path folder. It is not per instrument, not per dataset, not per RAW.

Code lens: why the reference is “global per folder”

```
% check_ref() always points to the same file inside raw_path
mat_file = fullfile(raw_path,'histone_layouts','0_ref_info.mat');
```

```
% If match found:
rt_ref = AllUnHis.rt_ref(ix);
ndebug = -1; % effectively: "use reference, do not re-learn"
```

Architecture note: in the codebase, ‘RT reference’ is not a single entity. EpiProfile mixes (i) a per-module hardcoded RT template, (ii) a folder-level persisted reference (0_ref_info.mat) that can impose RTs learned from other RAWs, and (iii) intra-run anchors derived from unmodified forms and/or intermediate outputs (*.xls) passed between modules. Figure 2.X summarizes where each layer enters and why reusing folders/outpaths can silently contaminate RT guidance.

Where RT 'reference' enters EpiProfile

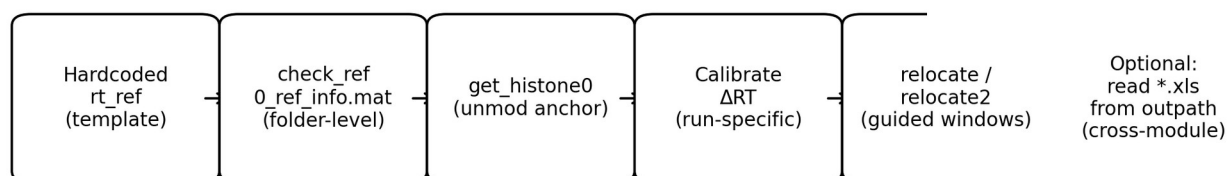


Figure 2.X. RT_ref layers in EpiProfile and persistence points (folder/outpath) that can induce cross-run references.

Layer 2 — intra-run anchoring + relocate()

Within a run, EpiProfile tries to detect the unmod peak with `get_histone0()`. If it succeeds, it shifts the entire RT template by a delta so all other peptideforms move together in that run.

Then `relocate()/relocate2()` often uses narrow RT windows around expected positions to find modified forms. This can be MS1-only or MS1+MS2-assisted depending on `special.nDAmode`.

WARNING: Key fragility

If unmod is missing or mislocalized, the delta shift is wrong or not applied. At that point, `relocate()` becomes a narrow-window search around the wrong place, which produces zeros without necessarily crashing.

Conservative operating policy (no code changes)

Goal: never apply a ZenoTOF-optimized RT reference to Orbitrap PXDs (or any heterogeneous LC setup) by accident.

The policy below is intentionally conservative: it sacrifices some convenience to minimize silent failure.

Golden rules (print these)

1. One folder (`raw_path`) = one homogeneous group (same LC method; ideally same instrument and close dates).
2. If you reuse a folder for a new dataset: delete `raw_path/histone_layouts/0_ref_info.mat` before running.

3. One output folder (outpath) = one RAW/run, or at least one homogeneous group. Do not share outputs across heterogeneous RAWs.
4. If in doubt about homogeneity: set `ndebug=2` (no reference) and run each RAW independently.

TIP: Practical directory template (recommended)

Create one top-level folder per dataset (e.g., per PXD). Inside it, create a subfolder per RAW. Run EpiProfile with `raw_path` pointing to each subfolder, so both `0_ref_info.mat` and any intermediate `*.xls` are naturally isolated.

```
# Example layout (one dataset with multiple RAWs)
PXD046034/
  RAW_001/ (contains RAW_001.raw)
  RAW_002/ (contains RAW_002.raw)
  ...
# Run EpiProfile separately per RAW folder (or per homogeneous batch)
```

Decision matrix: when to use `ndebug=0` vs `ndebug=2`

`ndebug` controls whether `check_ref()` is allowed to load `0_ref_info.mat` (reference RT).

```
% check_otherparas.m (defaults)
ndebug = 0; % 0: normal + reference RT
    % 1: debug
    % 2: normal but NO reference RT (per-file without 0_ref_info)
```

Use `ndebug=0` only when the RAWs inside `raw_path` are homogeneous (same LC method; ideally same instrument and close acquisition dates).

Use `ndebug=2` when `raw_path` contains heterogeneous RAWs (different instruments, different gradients, long time gaps, or unknown provenance).

Mass tolerance (`def_ptol`) — conservative guidance and failure modes

`def_ptol` is the ppm window used for MS1 peak extraction (and sometimes for MS2 helpers). It is a sensitivity/specificity knob.

Histone PTM peptideforms are dense in m/z space. Increasing `ptol` rescues weak targets but increases the chance of grabbing a neighbor peak (silent wrong quantification).

Baseline rule

- Start at 10 ppm (default).

- Increase to 20 ppm only if you have evidence of a systematic mass offset in your run (e.g., multiple strong targets appear consistently shifted).
- Avoid 30+ ppm unless your data are clearly not well-calibrated HRMS; treat it as a last resort and inspect plots/PSMs.

Common mass-related symptoms and conservative actions

Below are typical mass-accuracy failure modes and what they look like in outputs.

Mass failure M1 — systematic ppm offset (targets disappear at 10 ppm)

Symptom: many peptideforms are zero across the board, including strong expected ones, but RT plots show peaks slightly shifted in m/z .

Root cause: calibration drift, wrong conversion settings, or instrument-specific mass error profile.

Conservative fix: increase `ptol` to 20 ppm temporarily; validate by checking that multiple known strong peptides reappear at plausible RT and with consistent isotopic envelopes.

```
# Operational test
ptol = 10 ppm -> run -> count detected peptides
ptol = 20 ppm -> run -> do strong anchors reappear with clean XIC peaks?

# If only noisy/random peaks appear, revert: you are likely admitting wrong matches.
```

Mass failure M2 — false positives at high `ptol` (wrong peaks integrated)

Symptom: peptideforms become non-zero after increasing `ptol`, but shapes are irregular, RT is implausible (far from expected family), or replicate consistency collapses.

Root cause: wide m/z window collects unrelated ions in complex regions; histone peptideforms often have close neighbors.

Conservative fix: revert to 10 ppm; prioritize RT correctness (avoid reference leakage) and MS2 validation (`nDAmode`) rather than wide ppm.

RT failures: why “unmod zeros” happen (expanded catalog)

An unmod being zero is a red flag. unmod is the anchor; if it is missing, most other forms become unreliable even if they report numbers.

This section enumerates practical scenarios that lead to unmod zeros, plus conservative fixes (no code edits).

RT failure R1 — cross-dataset reference leakage via `0_ref_info.mat`

Symptom: unmod is consistently zero in a dataset where it should be abundant. Many other forms are also zero or look inconsistent.

Mechanism: `check_ref()` overwrites the `unmod RT_ref` with a value learned in a different LC context. Then `get_histone0` searches in a guided ± 5 min window around the wrong RT and finds no scans/peaks → returns zeros.

WARNING: Why this is worse than no reference

A bad reference can force a narrow-window search in the wrong place. With no reference, EpiProfile often searches globally and has a chance to recover the `unmod`.

5. Fix (preferred): isolate datasets into separate `raw_path` folders, delete `0_ref_info.mat`, keep `ndebug=0` within each homogeneous folder.
6. Fix (if you must keep mixed RAWs): set `ndebug=2` so `check_ref` is bypassed.
7. Verification: after rerun, check that `unmod RT` is >0 and that modified forms shift consistently (same delta).

RT failure R2 — outpath contamination between modules (H3_04v3_27_40 case)

Some modules read RT anchors from other modules' outputs (*.xls). Example: `H3_04v3_27_40` reads `H3_04_27_40.xls` to obtain a `ref_rts` array, then searches in ± 0.5 min windows around (`ref_rts - nshift`).

If outpath is shared across RAWs, the `H3_04_27_40.xls` used as reference may come from a different run/instrument, creating silent relocation to the wrong region.

8. Fix: make outpath unique per RAW or at least per homogeneous batch.
9. If you cannot: delete intermediate *.xls outputs before each run, or ensure deterministic ordering that prevents cross-run reuse (still risky).
10. Verification: check that `H3_04_27_40.xls` and `H3_04v3_27_40` outputs correspond to the same RAW.

```
% H3_04v3_27_40: secondary reference layer
out_file0 = fullfile(His.outpath,'H3_04_27_40.xls'); % reads [rt] block

% If this file is from another run, relocate() uses wrong windows.
```

RT failure R3 — guided window has no scans (edge-of-gradient / truncated run)

Symptom: `unmod` zeros primarily for late-eluting peptides; early peptides look fine.

Root cause: the run ends early, the gradient is shorter than assumed, or RT units/scales differ (e.g., seconds vs minutes in conversion). Window `t1/t2` may land outside available `MS1_index(:,2)` range.

Conservative fix: confirm RT scale (minutes) in `MS1_index`; ensure full gradient acquired; do not rely on `0_ref_info.mat` from longer runs.

RT failure R4 — drift too large for “small-shift” assumption

Symptom: unmod is found but many modified forms remain zero because relocate windows miss them. Alternatively, unmod is found at a plausible RT, but delta-shifted template still misaligns.

Root cause: the LC method differs enough that a simple global delta shift does not preserve relative ordering; some forms move nonlinearly (common when gradients are re-optimized).

Conservative fix: treat as a different homogeneous group; do not expect cross-dataset RT templates to transfer. Use ndebug=2 and rely on broader RT searches where available; consider running with MS2-assistance (nDAmode) if data permit.

MS2 / DDA mode and ambiguity failures

Histone peptideforms include isobaric combinations (e.g., acetyl+trimethyl vs propionyl+dimethyl). MS1-only quantification can produce plausible but incorrect assignments.

special.nDAmode controls whether MS2 evidence (PSMs) is used in parts of the pipeline.

Ambiguity A1 — MS1-only assigns the wrong isobaric form

Symptom: numbers exist (not zeros), but biological patterns are implausible (e.g., K27me3 appears where it should be absent; or mutually exclusive marks co-occur strongly).

Root cause: MS1-only extraction integrated a peak that corresponds to an isobaric neighbor.

Conservative fix: when possible, enable MS2/PSM extraction (nDAmode) and inspect PSM tables for the affected regions.

What the 4 stability patches do (and what they do not do)

You applied four robustness patches (get_histone11, get_rts, draw_layout, get_histone0) to prevent crashes when RT windows are empty or out of range.

These patches are essential for stability, but they do not make wrong references “safe”. They convert hard failures into soft failures (zeros + logs). That is good for batch processing, but it can hide silent misconfiguration unless you monitor QC.

- If you see many zeros after the patches, assume a configuration/reference issue first, not a MATLAB bug.
- Treat ‘unmod=0’ as an immediate stop signal for that peptide family; debug the RT and mass settings before trusting any downstream ratios.

Troubleshooting playbook: when you see many zeros

This is the conservative debug sequence that minimizes wasted time and prevents you from chasing the wrong layer.

Step 0 — classify the zero pattern

- Zeros for one peptide family only (e.g., only TKQTAR) → likely module-specific RT reference problem or local interference.
- Zeros for many families in the same RAW → likely mass tolerance, RT scale, or global reference leakage.
- Zeros only in some RAWs within the same dataset → folder mixing or outpath mixing; or per-run acquisition issues.

Step 1 — verify folder hygiene

11. Is there a histone_layouts/0_ref_info.mat in raw_path? If yes, is it from this dataset? If uncertain: delete it.
12. Is outpath shared across RAWs? If yes: isolate per RAW; delete intermediate *.xls files and rerun.
13. Rerun one RAW with ndebug=2 as a control. If zeros disappear, the reference layer is implicated.

Step 2 — verify RT scale and run completeness

14. Check MS1_index(:,2): is it in minutes and increasing to the expected end of gradient?
15. If end_rt is unexpectedly small (e.g., <20 for a 120-min method), suspect conversion/unit errors or truncated acquisition.
16. Confirm the RAW truly contains MS1 scans across the whole gradient.

Step 3 — test mass tolerance sensitivity

17. Run at 10 ppm, then 20 ppm. Compare peptide counts and inspect a few anchor peptides' XICs.
18. If 20 ppm rescues many peptides with clean peaks, consider keeping 20 ppm for this dataset.
19. If 20 ppm mainly adds noisy detections, revert and fix RT/reference instead.

Step 4 — validate with figures/PSMs for 1–2 representative peptides

Pick one abundant region (e.g., H4_4_17 acetylation family) and one problematic region (e.g., TKQTAR). Inspect the layout figures and, if enabled, PSM tables.

Do not trust matrix outputs until these checks pass.

Worked examples (realistic failure scenarios)

Example E1 — TKQTAR unmod becomes zero after moving to a new PXD

Context: TKQTAR module hardcodes unmod RT_ref≈18.34. A previous ZenoTOF run created 0_ref_info.mat with RT≈18 min. You then process an Orbitrap PXD where TKQTAR elutes at ~42 min due to a different gradient.

What happens: check_ref overwrites His.rt_ref(1) with ~18. get_histone0 uses a guided window and finds nothing → unmod=0. All modified forms are shifted from an unmod anchor that never existed → mostly zeros.

Fix: delete 0_ref_info.mat, isolate the Orbitrap dataset, rerun with ndebug=0 inside that dataset folder (new reference), or use ndebug=2 for per-run processing.

Example E2 — many peptideforms rescued by ptol=20 ppm but biology becomes nonsense

Context: a dataset shows mild mass drift (~12–15 ppm). At 10 ppm, many forms are missing. At 20 ppm, they appear, but patterns across replicates are inconsistent.

Interpretation: you admitted competing peaks in dense m/z regions. You rescued sensitivity but lost specificity.

Fix: revert to 10 ppm; fix calibration or conversion; use MS2/PSM validation for ambiguous regions.

Example E3 — H3_04v3_27_40 shows zeros only for some RAWs

Context: outpath is shared across RAWs. The first RAW produces H3_04_27_40.xls; subsequent RAWs reuse it as relocation anchor.

Symptom: H3_04v3_27_40 detections look fine for the first RAW and collapse (zeros) for later RAWs when their true RTs differ.

Fix: outpath per RAW; delete intermediate *.xls; rerun.

How to report an issue (WhatsApp / meeting template)

To debug quickly, every report should include a minimal reproducibility bundle:

- Dataset identifier (PXD), instrument, LC method if known (gradient length, column).
- The exact raw_path and outpath layout (are RAWs mixed? are outputs shared?).
- ndebug value, ptol value, nDAmode value.
- Which peptide module(s) are failing (e.g., H3_01_3_8, H3_04v3_27_40).
- One screenshot of the relevant layout plot (or the fact that the plot is empty).
- Whether unmod is zero and whether many families are zero.

TIP: Fast triage question

If you answer only one question first, answer this: “Is there a 0_ref_info.mat in this folder, and is it guaranteed to belong to this dataset?”

Appendix: code excerpts for orientation (read-only)

These snippets are included to help non-MATLAB users understand what the code is doing. They are not instructions to edit code.

```
% Typical unmod anchoring logic inside a peptide module
```

```
His.rt_unmod_orig = His.rt_ref(1);
```

```
[cur_rts,cur_intens,...] = get_histone0(...);
```

```
if cur_rts(1) > 0
```

```
    His.rt_ref(1) = cur_rts(1);
```

```
    delta = cur_rts(1) - His.rt_unmod_orig;
```

```
    His.rt_ref(2:end) = His.rt_ref(2:end) + delta;
```

```
end
```

```
% relocate() pattern: narrow-window search around expected RTs
```

```
delta = 0.5;    %  $\pm 0.5$  min window
```

```
t1 = ref_rts(hno) - nshift - delta;
```

```
t2 = ref_rts(hno) - nshift + delta;
```

```
[rts,top1_rt] = get_rts(..., t1, t2);
```

```
if isempty(rts)
```

```
    His.rt_ref(hno) = 0; % "no RT assigned" -> often yields zeros downstream
```

```
else
```

```
    His.rt_ref(hno) = top1_rt;
```

```
end
```

References

- [1] "An Introduction to Mass Spectrometry-Based Proteomics". Journal of Proteome Research. DOI: 10.1021/acs.jproteome.0c00592. (Provided background document).
- [2] "Comprehensive Overview of Bottom-Up Proteomics using Mass Spectrometry". Tutorial/manuscript (MeyerLab GitHub), 2024-05-07. (Provided background document).
- [3] Geshkovski, K.; Jaquemart, E.; Pradillo, M.; Gutierrez-Marcos, J. "Quantitative Profiling of Histone Variants and Posttranslational Modifications by Tandem Mass Spectrometry in Arabidopsis". Methods in Molecular Biology, 2025. (Provided background document).
- [4] "A Strategy to Characterize the Global Landscape of Histone Post-Translational Modifications within Tissues of Non-model Organisms". Journal of Proteome Research. DOI: 10.1021/acs.jproteome.0c00216. (Provided background document).

Appendix D. Minimal QC before moving to R (5–10 minutes)

Goal: quickly detect whether a run was compromised by (i) incorrect guided RT, (ii) overly strict mass tolerance, or (iii) signal/identification issues that generate matrices with many zeros.

Figures in this appendix are illustrative: what matters is the pattern, not the exact numeric values.

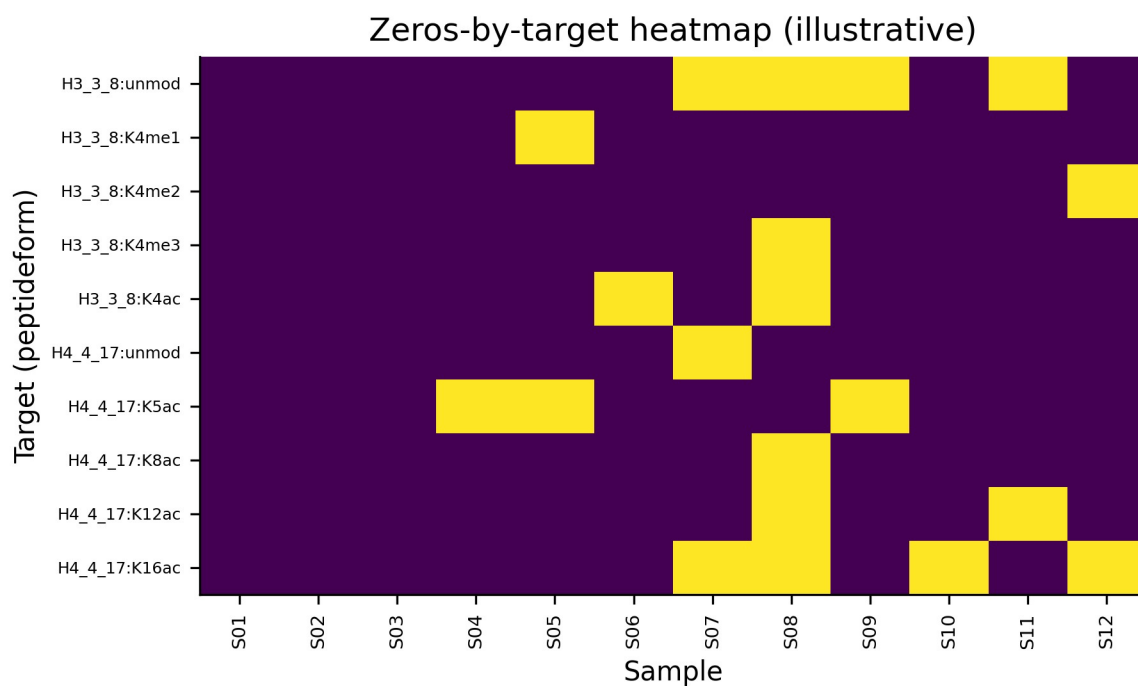


Figure D1 | Zeros-by-target heatmap. A dark vertical stripe suggests a run-wide problem (signal, RT, or preprocessing).

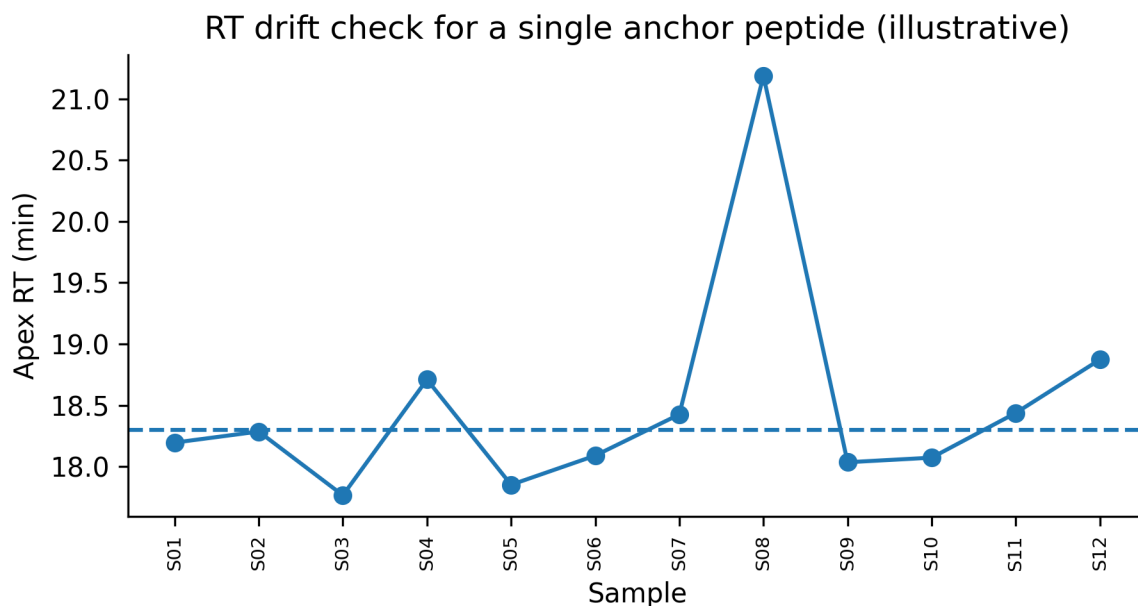


Figure D2 | RT drift of an anchor peptide (unmodified) across samples. Large outliers often correlate with guided-window failures.

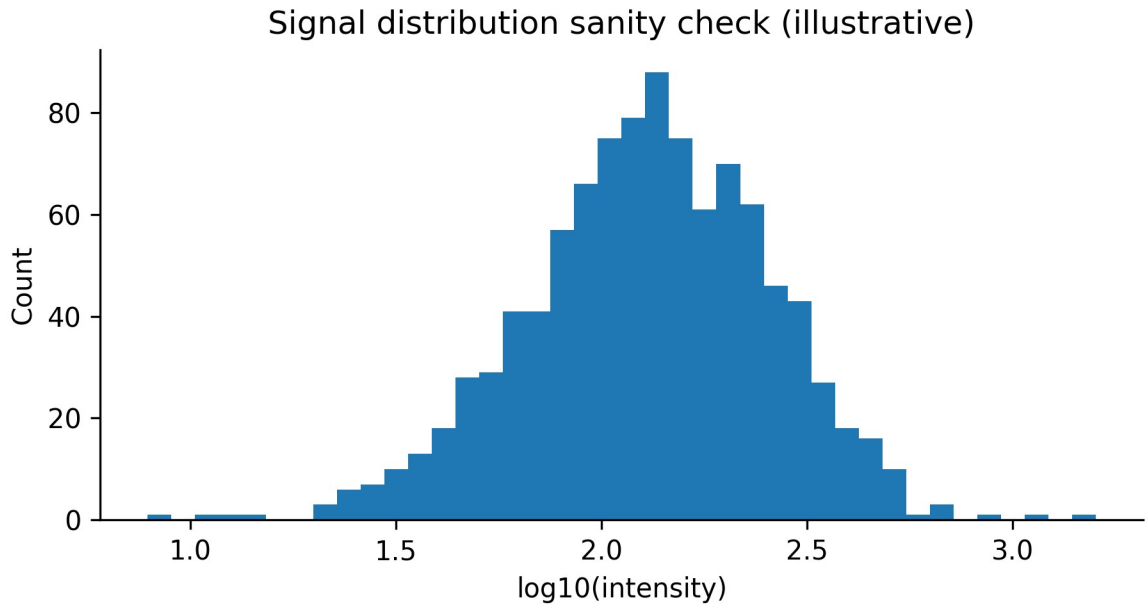


Figure D3 | Signal distribution (log10). Truncation or large shifts across runs typically indicate loading/LC differences or extraction failure.

Example minimal script (R) to quantify zeros and RTs from EpiProfile outputs:

```
# Requires: readr, dplyr, tidyr, ggplot2
x <- readr::read_tsv("histone_ratios.xls", skip=0) # note: this is TSV, not true Excel
# 1) fraction of zeros per sample
pct0 <- sapply(x[1:2], function(v) mean(as.numeric(v)==0, na.rm=TRUE))
print(sort(pct0, decreasing=TRUE)[1:10])

# 2) if RTs are present in the same file (the [rt] block): parse and plot RT of the unmodified form
# (see earlier sections for the Ratio/Area/RT block layout)
```